

Literature resolution of the sum-bifunction question

arXiv:1105.5550, Remark 7

Status. Literature already answered by a later general necessary-and-sufficient characterization. The application below is an inference; the later paper does not explicitly label its theorem as an answer to Remark 7.

The source question. Let X be reflexive, let $C \subset X$ be closed and convex, and let $F, G : C \times C \rightarrow \mathbb{R}$ be monotone, vanish on the diagonal, and have convex lower-semicontinuous second-variable slices. Under $0 \in \text{sqr}(C - C)$, Proposition 18 proves

$$A^F + A^G = A^{F+G}.$$

Remark 7 asks what additional hypothesis, no stronger than the preceding standard sufficient hypotheses, guarantees maximality of $F + G$.

Later answer. Hadjisavvas, Jacinto, and Martínez-Legaz prove in Theorem 2 of *Some Conditions for Maximal Monotonicity of Bifunctions* (published online 2015, DOI 10.1007/s11228-015-0343-6) that, for every monotone bifunction Q with convex lower-semicontinuous second-variable slices, maximal monotonicity is equivalent to three other conditions. One is pointwise maximality of the canonical support bifunction G_{AQ} . Another is the exact comparison condition

$$\begin{aligned} &\text{for every admissible PMMB } K \text{ with finite Fitzpatrick transform,} \\ &\exists x_K : \quad Q(x_K, y) + K(x_K, y) \geq 0 \quad (\forall y). \end{aligned} \tag{1}$$

Their fourth clause is an equivalent perturbed-equilibrium solvability condition for every $(x_0, x_0^*) \in X \times X^*$.

To apply the theorem, put $Q = F + G$ and extend it from $C \times C$ by

$$\widehat{Q}(x, y) = \begin{cases} Q(x, y), & x, y \in C, \\ +\infty, & x \in C, y \notin C, \\ -\infty, & x \notin C. \end{cases} \tag{2}$$

Then $\widehat{Q}$ is monotone, its second-variable slices are convex and lower semicontinuous, and $A^{\widehat{Q}} = A^Q$. Thus Theorem 2 applies directly. Conditions (1) and its equivalent clauses are exact additions to Proposition 18: they are necessary and sufficient, and hence cannot be stronger than any valid sufficient condition from the preceding remark.

Why something must be added. The original assumptions alone do not imply maximality. Take $X = C = \mathbb{R}$, $G = 0$, and

$$a(x) = \begin{cases} 0, & x \leq 0, \\ 1, & x > 0, \end{cases} \quad F(x, y) = a(x)(y - x).$$

The function a is nondecreasing, so F is monotone; every $F(x, \cdot)$ is affine and continuous, and $0 \in \text{sqr}(C - C)$. Nevertheless $A^F(x) = \{a(x)\}$ is not maximal: each $(0, t)$, $0 < t \leq 1$, is monotonically related to its graph but absent from it.

Scope. The 2015 theorem resolves the broad request by an exact characterization. It does not turn the answer into a single elementary semicontinuity assumption, so searches for especially simple sufficient conditions remain meaningful but are no longer the same open question.

Proof. Let $x \in C$. One has $\text{dom}(F(x, \cdot) + \delta_C) = \text{dom}(G(x, \cdot) + \delta_C) = \text{dom}((F + G)(x, \cdot) + \delta_C) = C$. By definition, $A^F(x) = \partial(F(x, \cdot) + \delta_C)(x)$. Note also that $(F(x, \cdot) + \delta_C) + (G(x, \cdot) + \delta_C) = (F + G)(x, \cdot) + \delta_C$. By [12, Theorem 2.8.7], the hypotheses imply

$$\partial(F(x, \cdot) + \delta_C)(x) + \partial(G(x, \cdot) + \delta_C)(x) = \partial(F(x, \cdot) + G(x, \cdot) + \delta_C)(x).$$

Consequently, $A^F(x) + A^G(x) = A^{F+G}(x)$ and since $x \in C$ was arbitrarily chosen, the conclusion follows. ■

Remark 7 Note that the hypotheses of Proposition 18 ensure that $\overline{\text{co}}h_{F+G}(x, x^*) \geq h_{F+G}^*(x^*, x) \geq c(x, x^*)$ for all $(x, x^*) \in X \times X^*$. Unfortunately, this is not enough in order to guarantee the maximality of $F + G$, which would follow for instance provided the *BO*-maximal monotonicity of this bifunction. However, checking also Remark 6, this additional assumption would make, at least in the reflexive case, the condition $0 \in \text{sqr}(C - C)$ redundant. Therefore, it remains as an open question what should one add to the hypotheses of Proposition 18 in order to obtain the maximality of $F + G$ under no stronger hypotheses than the ones in Remark 6.

Acknowledgements. The authors are grateful to their colleague E.R. Csetnek for valuable comments and suggestions on an earlier draft of the paper.

References

- [1] M. Ait Mansour, Z. Chbani, H. Riahi (2003): *Recession bifunction and solvability of non-coercive equilibrium problems*, Communications in Applied Analysis 7(2–3), 369–377.
- [2] E. Blum, W. Oettli (1994): *From optimization and variational inequalities to equilibrium problems*, The Mathematics Student 63(1–4), 123–145.
- [3] R.I. Boţ, E.R. Csetnek (2008): *An Application of the bivariate inf-convolution formula to enlargements of monotone operators*, Set-Valued Analysis 16(7–8), 983–997.
- [4] R.I. Boţ, S.-M. Grad, G. Wanka (2006): *Maximal monotonicity for the precomposition with a linear operator*, SIAM Journal on Optimization 17(4), 1239–1252.
- [5] R.I. Boţ, S.-M. Grad, G. Wanka (2007): *Weaker constraint qualifications in maximal monotonicity*, Numerical Functional Analysis and Optimization 28(1–2), 27–41.
- [6] R.S. Burachik, B.F. Svaiter (2002): *Maximal monotone operators, convex functions and a special family of enlargements*, Set-Valued Analysis 10(4), 297–316.
- [7] A.N. Iusem (2011): *On the maximal monotonicity of diagonal subdifferential operators* Journal of Convex Analysis 18(2), 489–503.
- [8] M. Marques Alves, B.F. Svaiter (2010): *Maximal monotonicity, conjugation and the duality product in non-reflexive Banach spaces*, Journal of Convex Analysis 17(2), 553–563.
- [9] N. Hadjisavvas, H. Khatibzadeh (2010): *Maximal monotonicity of bifunctions*, Optimization 59(2), 147–160.

$\inf_{y \in \mathbb{R} \setminus \{0\}} (x - y)(x^* + s - y) > -\infty$, for $s \in \{-1, 1\}$. The latter is obviously true since we have the infimum of a quadratic function in y with positive leading coefficient. Hence φ_B is finite-valued. For every $x \in \mathbb{R}$ we set $x' = x$. We see that statement 3) holds since $B(0) = [-1, 1]$ while $A(x)$ is one of the sets $\{-1\}$, $\{0\}$ and $\{1\}$. However, statement 1) does not hold.

The equivalence $1) \iff 3)$ in Proposition 3 generalizes the following characterization of maximality: a monotone operator A is maximally monotone if and only if for each $x \in X$, there exists $x' \in X$ such that $0 \in J(x' - x) + A(x')$. See for instance [6, p. 324].

3 Main Results

Our first main result gives conditions for a bifunction to be maximally monotone.

Theorem 2 *Let F be a monotone bifunction such that $F(x, \cdot)$ is convex and l.s.c., for each $x \in X$. Then the following statements are equivalent.*

- 1) F is an MMB.
- 2) G_{A^F} is a PMMB.
- 3) For every PMMB $H \in \mathcal{B}_m(X)$ such that Φ_H is finite-valued, there exists $x_H \in X$ such that

$$F(x_H, y) + H(x_H, y) \geq 0, \quad \forall y \in X.$$

- 4) There exist a finite-valued $H \in \mathcal{B}_s(X)$ and $p \in X$ such that

- (a) $H(p, \cdot)$ is continuous affine,
- (b) $H(p, z) + H(z, p) < 0$, $\forall z \in X \setminus \{p\}$,
- (c) For every $(x_0, x_0^*) \in X \times X^*$ there exists $\tilde{x} \in X$ satisfying

$$F(\tilde{x}, y) + H(x_0 - \tilde{x}, x_0 - y) - \langle y - \tilde{x}, x_0^* \rangle \geq 0, \quad \forall y \in X.$$

Proof $1) \Rightarrow 2)$ Let F be an MMB. If we set $F_1 = G_{A^F}$, then $F_1 \in \mathcal{B}_m(X)$. By Proposition 1, $A^{F_1} = A^F$, thus F_1 is an MMB. By Proposition 2, F_1 is a PMMB.

$2) \Rightarrow 3)$ Let H be as in statement 3). Setting again $F_1 = G_{A^F}$, by Proposition 2 both F_1 and H are MMB. Since $A^{F_1} = A^F$, the operators A^F and A^H are maximally monotone. In addition, $\varphi_{A^H} = \Phi_H$ is finite-valued.

Let B be the operator defined by $B(x) = -A^H(-x)$. It can be easily seen that B is maximally monotone, and $\varphi_B(x, x^*) = \varphi_{A^H}(-x, -x^*)$. Using implication a) $\Rightarrow$ b) of Theorem 1 we obtain that

$$(0, 0) = (w, w^*) + (v, v^*)$$

for some $(w, w^*) \in \text{Graph}(A^F)$ and $(v, v^*) \in \text{Graph}(-B)$. Then $w^* + v^* = 0$, with $w^* \in A^F(w)$, $v^* \in A^H(-v)$ and $w = -v$. Thus,

$$\begin{aligned} \langle y - w, w^* \rangle &\leq F(w, y), \quad \forall y \in X \\ \langle y - (-v), v^* \rangle &\leq H(-v, y), \quad \forall y \in X. \end{aligned}$$

Adding the two inequalities and setting $x_H = w$ we obtain

$$F(x_H, y) + H(x_H, y) \geq \langle y - x_H, w^* + v^* \rangle = 0, \quad \forall y \in X.$$

$3) \Rightarrow 4)$ We take $p = 0$ and $H = G_J$, where J is the duality mapping. We will show that H satisfies all conditions of 4).